

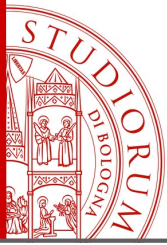
Ottimizzazione Matematica: Note su inversione matrici

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Richiami su inversione di matrici

- Data una matrice quadrata $A_{n \times n}$ essa è invertibile sse
 - le sue righe (colonne) sono tra loro linearmente indipendenti
 - il suo determinante è diverso da 0

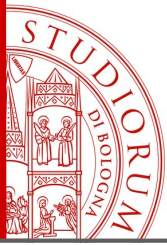
- Determinante matrice

- 2x2: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \det(A) = ad - cb$

- 3x3: metodo di Sarrus

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow \begin{array}{ccccc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array}$$

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$



Inversione matrici 2x2

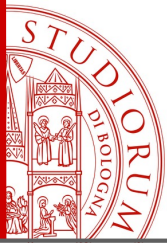
- data $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- Esempi:

$$\bullet A = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{2} \begin{bmatrix} -1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\bullet A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{2} \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$



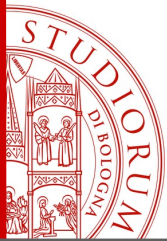
Inversione matrici 3x3

- $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

- $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} \text{cof}(a_{11}) & \text{cof}(a_{12}) & \text{cof}(a_{13}) \\ \text{cof}(a_{21}) & \text{cof}(a_{22}) & \text{cof}(a_{23}) \\ \text{cof}(a_{31}) & \text{cof}(a_{32}) & \text{cof}(a_{33}) \end{bmatrix}$

- dove $\text{cof}(a_{ij}) = (-1)^{i+j} C_{ij}$ e $C_{ij} = \det$ matrice ottenuta cancellando da A la riga i e la colonna j

Es: $C_{12} = (-1)^{1+2} \det \left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} \right) = -a_{21}a_{33} + a_{31}a_{23}$



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